MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 12000013

(ATMOSPHERIC CORRECTION)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-50785 comprise the complete requirements for the Atmospheric Correction Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50785	Circuit Card Assembly, Sight Current Generator, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

12000239	Sight Current Generator
10545455	Receptacle
12000013	Circuit Card Assembly, Atmospheric Correction

PACKAGING DATA SHEET

12000013	Packaging of Circuit Card Assembly, Atmospheric Condition

(Copies of specifications, standards, drawings, and publications required by Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Item definition.- This assembly is a single ended printed circuit board consisting of one capacitor, one set of resistors and an operational amplifier. The assembly is used to provide scaling factors within the Sight Current Generator Drawing Number 12000239 for air density characteristics.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 12000013.

3.1.2 Government furnished property list. Not applicable.

3.2 Characteristics.

3.2.1 Performance.- The assembly shall be capable of performing the following requirements when assembled in accordance with Drawing Number 12000013.

a. Provide ballistic scaling for the Sight Current Generator Air Density Circuit.

b. Provide a precision voltage source for the Sight Current Generator Air Density Circuit.

3.2.1.1 Input and output levels.

3.2.1.1.1 Input power.- Input power shall be supplied to the assembly as specified in Table I.

TABLE I - INPUT VOLTAGES

Pin	Voltage
4	+15 Vdc ± 0.1 percent
14	-15 Vdc ± 0.1 percent
28	Ground

3.2.1.1.2 Output levels.- The requirements specified in 3.2.1.1.2.1 shall be met when supplied with the specified voltage input.

3.2.1.1.2.1 Pin 17 output.- The output between pin 17 and pin 28 shall be -3.938 Vdc ± 1.0 percent.

3.2.1.2 Resistance and capacitance values.- The resistance and capacitance values shall be as specified in Table II. The maximum and minimum values shall be derated at temperature extremes of -40°F and +160°F per the applicable MIL specification covering each type resistor.

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TABLE II - RESISTANCE AND CAPACITANCE VALUES

Between		Minimum Reading		Maximum Reading	
Pin	Pin	Ohm	Microfarad	Ohm	Microfarad
15	TP2	2,090		2,310	
15	16	0.0000		0.0000	
13	18	0.0000		0.0000	
18	12	69,630		69,769	
12	11	87,112		87,287	
11	10	110,889		111,111	
10	9	147,852		148,148	
9	8	204,795		205,205	
8	7	283,716		284,284	
7	6	449,550		450,450	
6	5	774,225		775,775	
5	3	1,520,000		1,680,000	
3	2	4,845,000		5,355,000	
17	28		0.09		0.11
28	TP1	0.0000		0.0000	

3.2.1.3 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-50785.

3.2.2 Physical characteristic.- The physical characteristics shall be as specified in drawing 12000013 and the engineering parts list.

3.3 Design and construction.

3.3.1 Production drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-50785, drawing 12000013 and associated lists.

3.3.2 Standards of manufacture.- The standards of manufacture shall be those specified in specification MIL-C-50785.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-50785.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-50785.

4.1.2 Special tests and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-50785.

4.2 Quality conformance inspections.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-50785.

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4.2.1 Test equipment.

4.2.1.1 Power supply (PS1).- The power supply shall be +15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.2 Power supply (PS2).- The power supply shall be -15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.3 Digital ohmmeter.- The digital ohmmeter shall have a resistance readout to five digits from 100 ohms to 10,000,000 ohms with 0.01 percent accuracy.

4.2.1.4 Digital voltmeter (DVM).- The digital voltmeter shall have an accuracy and resolution of .01 percent or better from zero to 100 Vdc. Input impedance shall be greater than 10 megohms on all voltage scales.

4.2.1.5 Capacitance bridge.- The capacitance bridge shall have a capacitance readout from 0.001 microfarad to 10 microfarads with one percent or better, accuracy and resolution.

4.2.2 Functional test.

4.2.2.1 Test connections.- Power supplies PS1 and PS2 shall be connected to the assembly under test in accordance with Figure 1 using receptacle 10545455.

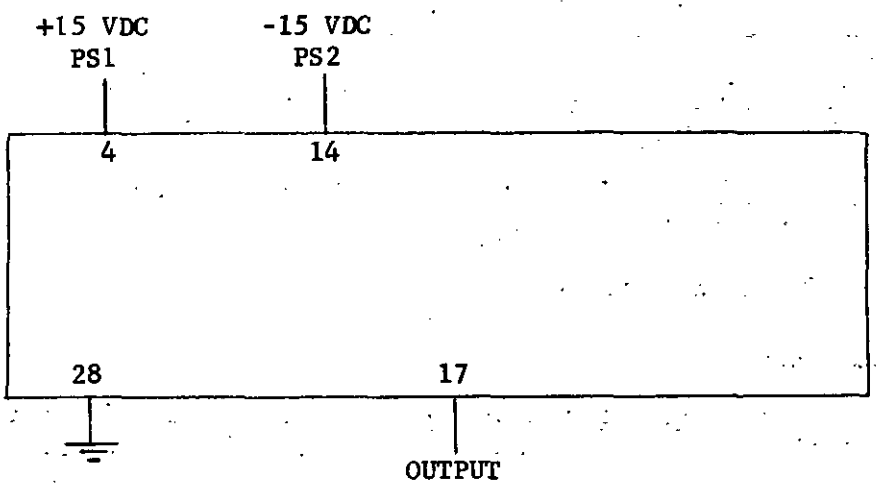


Figure 1. Test Connections

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4.2.2.2 Test procedure.- Compliance with the requirements of 3.2.1.1.2 through 3.2.1.2 shall be verified as follows:

a. With inputs specified in Table I, the voltage measured with the DVM between pin 17 and pin 28 (ground) shall be -3.938 Vdc ± 1.0 percent.

b. With no voltages applied, the resistance measured with the digital ohmmeter between each set of pins shall be as specified in paragraph 3.2.1.2.

c. With no voltages applied, the capacitance measured with the capacitance bridge between each set of pins shall be as specified in paragraph 3.2.1.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the level A requirements of Packaging Data Sheet (PDS), AMC Form 1029, 12000013.

6. NOTES

6.1 The notes specified in the general specification MIL-C-50785 are applicable to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1285-A432

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-50785/13(MU) Circuit Card Assembly: 12000013 (Atmospheric Correction)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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